COVER LETTER

Dear Editors of *Journal of Electronic Testing*,

Dear Editor,

Please find attached our manuscript entitled *"Experimental Error Modeling and Post-Processing Correction of NanoVNA Measurements Using Low-Cost Calibration Standards"* for consideration for publication in the Journal of Electronic Testing.

This work addresses the measurement accuracy limitations of low-cost vector network analyzers, specifically the NanoVNA, which has become widely used in educational, experimental, and resource-constrained RF environments due to its affordability and accessibility. The manuscript proposes an experimental error modeling and post-processing correction framework based on low-cost short-open-load calibration standards and open-source software tools, with the goal of improving S-parameter measurement reliability without requiring hardware modifications or metrology-grade calibration kits.

The main contribution of the study is a frequency-dependent one-port error modeling methodology that estimates directivity, source match, and reflection tracking terms through complex least-squares fitting of calibration measurements exported in Touchstone format. These estimated coefficients are then used to perform an external numerical correction of measured S-parameters, enabling a practical enhancement of NanoVNA measurement performance under realistic low-cost conditions. The proposed method is experimentally validated using representative RF devices, including mismatched loads, attenuators, and LC resonant circuits, showing consistent reductions in mean absolute error, root mean square error, and high-percentile error metrics after correction.

We believe this manuscript is well suited to the scope of the Journal of Electronic Testing because it combines experimental characterization, error modeling, calibration-related analysis, and validation on practical electronic test scenarios. The study is especially relevant for readers interested in low-cost instrumentation, test accuracy enhancement, RF measurement methodologies, and accessible alternatives for laboratory and educational environments.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no competing interests, and no external funding was received for this research. The raw dataset and analysis scripts supporting the study are publicly available through Zenodo, as indicated in the manuscript.

Thank you for your time and consideration. We would be grateful for the opportunity to have this work evaluated for publication in the J*ournal of Electronic Testing.*

Telmo Miguel-Medina